Name _____

Student Number _____

Tutorial 5 Worksheet: Virus Overview and Life Cycle

Part A: Fill in Viral Sleuthing Table OR replace with a photo of your memory aid

Virus	Measles	Influenza A	HCV
Genus	Morbillivirus	Alphainfluenzavirus	Hepacivirus
Family	Paramyxoviridae	Orthomyxoviridae	Flaviviridae
Structure	Helical nucleocapsid	Helical RNPs	Icosahedral capsid
Envelope?	Yes	Yes	Yes
Genome 1. Genetic Type: 2. Baltimore Classification: 3. Segmented Y/N:	1. (-) ssRNA 2. V 3. N	1. (-) ssRNA 2. V 3. Yes, 8 segments	1. (+) ssRNA 2. IV 3. No
Viral Genes and Proteins	N (nucleocapsid) P (polymerase cofactor), L (RdRP), M (matrix), H (attachment), F (fusion), V/C (immune evasion)	PB1/PB2/PA (polymerase), HA (attachment/fusion), NA (release), NP (RNA binding), M1 (matrix), M2 (ion channel), NS1 (IFN inhibition), NEP (nuclear export)	Core (capsid), E1/E2 (entry), p7 (ion channel), NS2/NS3 (proteases), NS4A (cofactor), NS4B (membrane remodeling), NS5A (replication/assembly), NS5B (RdRP)
Tissue Tropism	Immune cells, respiratory epithelial cells	Respiratory epithelial cells	Hepatocytes (liver cells)
Viral Attachment Protein	H protein	HA (hemagglutinin)	E1/E2 glycoproteins
Host Receptor	CD150 (SLAM), Nectin-4	Sialic acid (α2,6 or α2,3)	CD81, SR-BI, CLDN1, OCLN
Site of replication	Cytoplasm	Nucleus (RNA replication), cytoplasm (translation)	Cytoplasm (ER-associated membranes)
Transmission	Airborne respiratory particles	Respiratory droplets and aerosols	Blood-to-blood contact
Outcomes of infection	Acute infection, lifelong immunity, rare SSPE	Acute infection, recovery, sometimes severe pneumonia	Often chronic infection, cirrhosis, liver cancer

Part B: Draw and annotate the life cycle of Measles, Inf A, or HCV (for practice we recommend doing all three!)

Accessibility Option:
You may represent the viral lifecycle for one virus in **any of the following formats**:

- A **drawing or annotated diagram** (hand-drawn or digital).
- A **written description or step-by-step outline** of the lifecycle.
- A **short audio or voice recording** (≤2 min) explaining the lifecycle stages.

Please see the other file submitted together

① Attachment
H binds host receptor
- CD150 (SLAM) on immune cells
- Nectin-4 on epithelial cells

CD150 (SLAM)
Nectin-4

② Fusion and entry
F mediates membrane fusion at the plasma membrane (Direct fusion, not endocytosis)

③ Uncoating
RNP remains intact in cytoplasm (genome stays coated by N proteins)

L RNP released into cytoplasm

④ Primary transcription
L+P transcribe (-)RNA genome into viral mRNAs (in cytoplasm)

Viral RNA

Gene order (3'-5'):
3'-N-P/V/C-M-F-H-L-5'

⑤ Translation of viral proteins
cytoplasmic proteins
(translated on free ribosomes)
· N
· P
· L
· M
· V/C

Free ribosomes

H, F (synthesized as F_0, cleaved to active F)

rough ER
Golgi

Membrane glycoproteins
(synthesized in rough ER, processed in ER/Golgi)

H & F trasported to plasma membrane

cytosol

⑥ Genome replication
When enough N protein accumulates, plymerase switches to replication.

(-) genome → (+) antigenome (full-length) → New (-) genome (encapsidated by N protein)

⑦ Assembly
Nucleocapsid interacts with M protein at the plasma membrane. H & F are already in the membrane

M

⑧ Budding & release
New virions bud from plasma membrane, acquiring envelope with H and F

Released virion can infect neighboring cells

⑨ Cell-cell fusion (syncytia)
F protein can also cause fusion of infected cell with neighboring cells, forming syncytia (multinucleated giant cells), which promotes local spread without extracellular release.

Mo Tu We Th Fr Sa Su

Memo No. ____
Date / /